

Heuristic Analysis of Cross-Base Palindromic Primes

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1. Introduction:

We study the sequence of integers $n \geq 2$ satisfying four simultaneous conditions:

1. n is prime;
2. n is a palindrome in base 10;
3. The binary representation of n , interpreted as a base-10 integer, is prime;
4. The binary representation of n , interpreted as a base-10 integer, is a palindrome in base 10.

An exhaustive computational search up to 10^{14} using a palindrome generator combined with the deterministic Miller-Rabin primality test found exactly four terms:

$$S = \{3, 5, 313, 7284717174827\}$$

In this document, we present a heuristic argument suggesting that this sequence is finite, and that no additional terms are expected beyond these four values.

2. Individual Probabilities

We estimate the probability that a large integer n satisfies each of the four conditions independently.

2.1. Probability of being prime

By the Prime Number Theorem, the probability that a random integer n is prime is:

$$P_1(n) \approx \frac{1}{\ln(n)}$$

2.2. Probability of being a palindrome in base 10

A number with d decimal digits is a palindrome if the second half of its digits is a mirror of the first half. The number of palindromes with d digits is $9 \times 10^{\lceil d/2 \rceil - 1}$, while the total number of d -digit numbers is $9 \times 10^{d-1}$. Therefore:

$$P_2(n) \approx \frac{1}{\sqrt{n}}$$

2.3. Probability of the binary representation being a palindrome

The binary representation of n has $b \approx \log_2(n)$ bits. Since n is odd (all primes greater than 2 are odd), the first and last bits are both 1 (they automatically match). The remaining $\lfloor b/2 \rfloor - 1$ pairs of bits must match, each with probability $1/2$. This gives:

$$P_3(n) \approx \left(\frac{1}{2}\right)^{\frac{b}{2}-1} = \frac{2}{\sqrt{2^b}} \approx \frac{\sqrt{2}}{\sqrt{n}}$$

2.4. Probability of the binary (read as decimal) being prime

The binary representation of n , when read as a base-10 integer, has approximately $b \approx \log_2(n)$ decimal digits. Its value is of the order 10^b . By the Prime Number Theorem:

$$P_4(n) \approx \frac{1}{\ln(10^b)} = \frac{1}{b \cdot \ln(10)} = \frac{1}{\log_2(n) \cdot \ln(10)}$$

3. The Combined Probability

Assuming asymptotic independence of digit representations in multiplicatively independent bases (a standard heuristic related to Furstenberg's conjecture and the normality conjecture), the combined probability is the product:

$$P(n) = P_1(n) \cdot P_2(n) \cdot P_3(n) \cdot P_4(n)$$

Substituting:

$$P(n) \approx \frac{1}{\ln(n)} \cdot \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{2}}{\sqrt{n}} \cdot \frac{1}{\log_2(n) \cdot \ln(10)}$$

Simplifying $\sqrt{n} \cdot \sqrt{n} = n$:

$$P(n) \approx \frac{\sqrt{2}}{n \cdot \ln(n) \cdot \log_2(n) \cdot \ln(10)}$$

Using the change-of-base formula $\log_2(n) = \frac{\ln(n)}{\ln(2)}$:

$$P(n) \approx \frac{\sqrt{2} \cdot \ln(2)}{n \cdot \ln^2(n) \cdot \ln(10)}$$

We define the constant:

$$C = \frac{\sqrt{2} \cdot \ln(2)}{\ln(10)} \approx \frac{1.4142 \times 0.6931}{2.3026} \approx 0.426$$

Thus:

$$\boxed{P(n) \approx \frac{C}{n \cdot \ln^2(n)}}$$

4. The Integral and the Expected Count

To estimate the expected number of terms up to a bound N , we integrate the probability:

$$E(N) = \int_2^N P(n) dn = C \int_2^N \frac{dn}{n \cdot \ln^2(n)}$$

4.1. Solving the integral

We use the substitution $u = \ln(n)$, so $du = \frac{1}{n} dn$:

$$\int \frac{dn}{n \cdot \ln^2(n)} = \int \frac{du}{u^2} = -\frac{1}{u} = -\frac{1}{\ln(n)}$$

Applying the limits from 2 to N :

$$E(N) = C \left[-\frac{1}{\ln(n)} \right]_2^N = C \left(-\frac{1}{\ln(N)} + \frac{1}{\ln(2)} \right)$$

$$E(N) = C \left(\frac{1}{\ln(2)} - \frac{1}{\ln(N)} \right)$$

4.2. The limit as $N \rightarrow \infty$

As $N \rightarrow \infty$, $\ln(N) \rightarrow \infty$, so $\frac{1}{\ln(N)} \rightarrow 0$. Therefore:

$$E(\infty) = \frac{C}{\ln(2)} \approx \frac{0.426}{0.693} \approx 0.615$$

The integral converges to a finite value. This is strong heuristic evidence that the sequence is finite.

5. Computational Method

The search was performed using:

1. A **palindrome generator** that constructs only base-10 palindromes (avoiding brute-force iteration over all integers).
2. The **deterministic Miller-Rabin primality test** with bases $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$, which is exact for all $n < 2^{64}$.
3. A **binary palindrome filter** using string reversal.

4. The **Rule of 3 filters** to eliminate candidates where the binary has a number of 1s divisible by 3.

The algorithm runs in $O(1)$ memory and processes approximately 10^6 candidates per second on a standard desktop computer.

6. Relationship to A046472

The sequence A046472 (Primes that are palindromic in bases 2 and 10) contains the terms:

3, 5, 7, 313, 7284717174827, 390714505091666190505417093

Our sequence is a **subsequence** of A046472, containing only those terms whose binary representation, when read as a base-10 integer, is also prime.

For example:

- 7 is in A046472, but its binary is 111, and $111 = 3 \times 37$ is not prime.
- The 6th term of A046472 has an 89-bit binary palindrome with exactly 36 ones. Since 36 is divisible by 3, the binary read as decimal is divisible by 3 and therefore not prime.

This "Rule of 3" filter (if the count of 1s in the binary is divisible by 3, the number is divisible by 3) provides an elegant and efficient way to eliminate candidates without expensive primality testing.

References

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